1.4.1 ▶ ▶ ▶ ▶ Angular Quantities Are the Same About Any Pivot

Consider a rigid body rotating freely with angular velocity $\boldsymbol{\omega}$ about an arbitrary point 1 on the body. Then, consider another point 2 a distance r away from the point of rotation. By the definition of rigid body rotation, this point has a speed $v = r\boldsymbol{\omega}$ in the direction shown in the diagram.

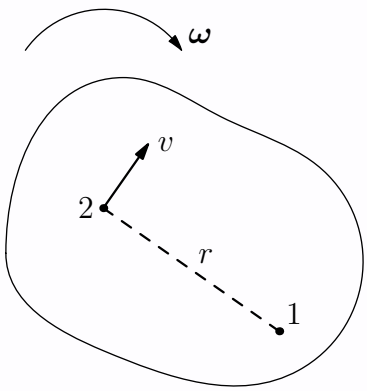

Fig. 1.7

Transforming into the reference frame of 2, we can find the angular velocity of the object about this point. Since it's a rigid body, the angular velocity of the body at 2 is equal the speed of 1 in 2's reference frame divided by r. Since 1 and 2 move relative to each other, this velocity is simply v. Hence,

$$\begin{aligned}\boldsymbol{\omega}_2 &= v/r \\ &= \frac{r\boldsymbol{\omega}}{r} \\ &= \boldsymbol{\omega}\end{aligned}$$

Thus, the angular velocity about any point on a rigid body is the same. It also follows that other angular quantities are the same, such as angular acceleration. This result is important, since the angular acceleration of a rigid body can be found by taking the torque about any fixed point, an important fact if some axes are easier to take torque around ★